Extra Problems

Please work on the problems below and then discuss the problems at the Week 4 Extra Problems Discussion forum.

Extra problems week 4

For our purposes here a number is a game x with x = p/2^n, or x = -p/2^n. The positive ones are defined recursively by

(2 p + 1)/2^(n+1) = {p/2^n | (p+1)/2^n},

with

The case n=0 and n positive integer can be defined separately by 0 = { | }, and p + 1 = {p | }.

1. Show directly that x = {1/16 | 3/4} is not the average of 1/16 and 3/4, namely y = 5/16 = {1/4 | 3/8} by showing a winning strategy for someone ((L or R) going first in x-y.
2. Show that {* | 5} is a number. Find it
3. If x is a number (for our purposes p/2^n, or its negative), then x + * = {x | x}. This is a special case of the Number Avoidance Theorem: If x is a number and G is not, then the best play in G + x is to an option of G.
4. Show that if x is a negative number, then Tiny(x) = {0 | (0 | -x }} is a number.
5. (For Mathematicians) Let omega= x = {0,1,2,3,... |} Find x+1, x+2, etc., Find 2x, 2x, 3x, 4x, etc. Can you guess what x . x would be? All of these can be drawn (sort of) as infinite Hackenbush games. What do they look like?